INFIX TO POSTFIX USING STACK

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
#define MAX 100

char stack[MAX];
int top = -1;

void push(char ch){
   stack[++top] = ch;
}

char pop(){
   return stack[top--];
}

char peek(){
   return stack[top];
}

int precedence(char ch){
   if (ch == '^')
      return 3;
   else if (ch == '*' || ch == '/')
      return 2;
```

```
    else if (ch == '+' || ch == '-')
        return 1;
    else
        return 0;
}

void infixToPostfix(char infix[], char postfix[]){
    int i, j = 0;
    char ch;

    for (i = 0; infix[i] != '\0'; i++) {
        ch = infix[i];

        if (isalnum(ch)) {
            postfix[j++] = ch;
        }
        else if (ch == '('){
            push(ch);
        }
        else if (ch == ')')  {
            while (top != -1 && peek() != '('
                postfix[j++] = pop();

            if (top != -1)
                pop();
        }
        else {
            while (top != -1 &&
```

```
            precedence(peek()) >= precedence(ch))  {
          postfix[j++] = pop();
        }

        push(ch);
      }
    }

    while (top != -1)
      postfix[j++] = pop();

    postfix[j] = '\0';
}

int main(){
    char infix[MAX], postfix[MAX];

    printf("========================================\n");
    printf("    INFIX TO POSTFIX CONVERTER\n");
    printf("========================================\n");

    printf("Enter infix expression: ");
    scanf("%s", infix);

    infixToPostfix(infix, postfix);

    printf("Postfix expression: %s\n", postfix);
```

```
    return 0;
}
```

OUTPUT

